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B.Tech. Degree IV Semester Examination April 2019

ME 15-1404 APPLIED THERMODYNAMICS (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) State Clausius statement of the second law of thermodynamics. Is it possible to intersect two reversible adiabatic paths? Justify your answer.
- (b) Define increase in entropy principle. Using this show that actual work is less than the reversible work.
- (c) Define the concept of availability. Determine the available energy of 100m³ of complete vacuum. Assume dead state as 100kPa and 300K.
- (d) Define the following (i) pure substance (ii) Triple point (iii) Critical point
- (e) A rigid container is filled with steam at 600kPa and 200°C. At what temperature does the steam begin to condense when it is cooled? Also determine the corresponding pressure.
- (f) Draw T-S diagram of Rankine cycle using dry saturated steam and develop the equation for Rankine cycle efficiency.
- (g) Explain with a suitable sketch the effect of back pressure on a Convergent-Divergent nozzle.
- (h) What is Wilson line? Explain its significance.
- (i) Clearly distinguish between ideal and real gases. Mention any two equations of state you know off.
- (j) A gas mixture consists of 6kmol of H₂ and 4kmol of N₂. Determine the mass of each gas and the gas constant of the mixture.

PART B

(4 × 10 = 40)

- II. (a) Two Carnot engines A and B are connected in series between two thermal reservoirs maintained at 1000K and 100K respectively. Engine A receives 1680 kJ of heat from the high temperature reservoir and rejects heat to a reservoir at temperature T. The Carnot engine B receives heat from this reservoir and reject heat to low temperature reservoir (100K). If engines A and B have equal thermal efficiencies, determine (i) The temperature at which heat is rejected by engine A (ii) Work done by the engine A and B (iii) The heat rejected by the engine B (iv) If engines A and B deliver equal work, find the efficiencies. (6)
 - (b) Check the possibility of the scheme of extracting 400kJ/kg of work output from a reversible heat engine embedded between two heat sources – one is a finite body at 900K and the other is a heat reservoir at 300K. Given the heat capacity of the body is 1.048kJ/kgK. (4)
- OR**
- III. (a) A reversible heat engine is used to drive a reversible heat pump. The heat engine takes in Q₁ heat units at T₁ and rejects Q₂ at T₂. The heat pump abstracts Q₄ from the sink at T₄ and discharges Q₃ at T₃. Develop an expression for the ratio of Q₄/Q₁, in terms of the four temperatures. (5)
 - (b) A battery is used to start a petrol car. 5 MJ of electrical energy is required to start the car. In order to get the same energy from compressed air of 8MPa and 27°C, what will be the volume of the container of the compressor? Assume dead state as 1bar and 27°C. (5)

(P.T.O)

- IV. (a) What is the meaning of wet, dry saturated and superheated steam? Discuss the graphical representation of T-v diagram for generation of superheated steam from -10°C of ice. (5)
- (b) Steam at 15 bar and 300°C expands isentropically in a steam turbine till the temperature falls to 80°C . Find the condition of steam at the end of expansion process and the work done per kg of steam. If the steam flow rate is 10kg/sec, what power will be produced by the turbine? (5)

OR

- V. (a) State the essential difference between Carnot and Rankine steam engine cycles. Sketch the Rankine cycle on p-v and T-s plane. (4)
- (b) A steam turbine working on Rankine cycle is supplied with dry saturated steam at 25bar and the exhaust take place at 0.2bar. For a steam flow rate of 10kg/sec, determine: (i) quality of steam at the end of expansion (ii) Turbine shaft work (iii) power required to drive the pump (iv) work ratio (v) Rankine efficiency (vi) heat flow in the condenser. (6)
- VI. (a) Define degree of reaction. Show that for Parson's reaction turbine the degree of reaction is 0.5. (4)
- (b) Steam expands through a Convergent- Divergent nozzle from 5 bar and dry saturated to a back pressure of 0.2 bar. Mass flow is 2kg/sec. Calculate the exit and throat areas when the friction loss in the divergent part is 10% of the total heat drop. Also determine the quality of steam at the inlet of the condenser. Neglect approach velocity and assume the expansion index is 1.135. (6)

OR

- VII. (a) Derive an expression for maximum flow rate of steam through a steam nozzle. Indicate the assumptions used. (4)
- (b) The rotational speed of an impulse turbine wheel is 3000rev/min. The nozzle is inclined at 20° to the plane of the wheel and its efficiency is 0.89. The isentropic heat drop for the stage is 159kJ/kg. If the ratio of blade speed to steam speed is 0.4, the blade velocity coefficient is 0.82 and the blading efficiency is 0.76. Find (i) the mean blade ring diameter (ii) the residual energy of steam at outlet per kg of steam (iii) the power developed by the wheel when the steam flow is 15kg/sec and (iv) the blade angles. (6)
- VIII. (a) State the Dalton's law of partial pressures. On what assumptions this law is based? (3)
- (b) Explain Gravimetric and volumetric analysis and make a relation between the two analyses. (3)
- (c) A mixture of 1kg of O_2 and 3kg of N_2 at 30°C and 10bar is compressed in a reversible adiabatic process to 60 bar. Calculate (i) The final partial pressure of the constituents (ii) The final temperature (iii) The change in internal energy of the mixture. Given C_v and C_p of N_2 are 0.74 kJ/kg and 3.5kJ/kg and C_v and C_p of O_2 are 3.5 kJ/kg and 5.5kJ/kg respectively. (4)

OR

- IX. (a) Prove the following (i) $R_{mix} = R_{mol} \sum \left[\frac{x_i}{M_i} \right]$ (ii) $M_{mix} = \sum \left[\frac{1}{\frac{x_i}{M_i}} \right]$ (4)
- (b) State Gibbs-Dalton law. (2)
- (c) Two vessels are connected by a valve. One vessel contains 0.6kg mole of nitrogen gas at 50°C and 15bar pressure. The other vessel holds 3.5kg of the same gas at 25°C and 6.5bar. The valve is opened and the gases are allowed to mix while having heat interactions with the surroundings. If the final equilibrium temperature is 30°C , (i) calculate the final equilibrium pressure and the amount of heat transferred to the surroundings, (ii) considering the vessel to be perfectly insulated, find the final temperature and pressure which would have been reached. Take $\gamma = 1.4$. (4)
